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ORIGINAL RESEARCH

Electrophysiological Changes in the Peripheral Nervous System After Subacute Spinal Cord Injury

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Abstract

Objective: To assess factors affecting electrophysiological changes in the peripheral nervous system below the neurologic level of injury (NLI) in patients with subacute spinal cord injury (SCI).

Design: Retrospective observational study.

Setting: An inpatient rehabilitation center of a university hospital.

Participants: Through reviewing the medical records of 151 subjects with SCI, 42 without any other disease inducing peripheral neurologic abnormalities were included. They were classified into 2 groups, with or without denervation potentials in electromyography (EMG) below NLI.

Intervention: Not applicable.

Main Outcome Measures: Demographics and clinical characteristics including NLI, American Spinal Injury Association Impairment Scale (AIS), and Lower Extremity Motor Score were compared. Results of electrophysiological study including nerve conduction study, somatosensory-evoked potential (SSEP), and motor-evoked potential (MEP) were compared.

Results: Denervation potentials in EMG below NLI were observed in 20 subjects, and 10 of them were AIS A or B, but there was none in subjects without denervation potentials ($P<.001$). The lower extremity motor score was 4.35±7.74 in the group with denervation potentials, lower than 33.64±13.60 of the opposite group ($P<.001$). In the analysis of electrophysiological study, patients with denervation potentials showed a higher proportion of no response than patients without denervation potentials (60.0% vs 11.4% in peroneal nerve conduction study, 35.0% vs 2.3% in tibial nerve conduction study, 80.0% vs 18.2% in SSEP, 87.5% vs 22.7% in MEP; $P<.001$, respectively). Additionally, greater axonal loss, based on decrease of amplitude without delayed latency on nerve conduction study, was observed in the group with denervation potentials than the opposite group ($P<.001$).

Conclusion: Among subjects with subacute SCI, cases of peripheral nervous dysfunction below the injury site occur, possibly associated with the severity of SCI.

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Given the general opinion that the motor neuron below neurologic level of injury (NLI) is anatomically intact, including the peripheral nerve axon, several studies have been published on the stimulation of motor neurons to induce functional improvement in patients with spinal cord injury (SCI).[1,2] However, a dysfunction of the peripheral nervous system below the lesion following SCI or the brain motor cortex has been reported.[3,4] This indicates that the central nervous system and the peripheral nervous system are closely related.

Distal atrophy of the spinal cord below the injury site is known to occur by Wallerian degeneration.[5] However, abnormal spontaneous activity (SA) that was not directly related to SCI on the muscles of the distal limbs were reported in electrophysiological studies in some patients with SCI.[6-9] Although abnormal SA frequently disappears over time, some studies have reported that abnormal SA sustained in the chronic phase of SCI can last for >1 year. In the early era of studies associated with SCI, these phenomena were considered to arise as a result of dysfunction of motor neurons during the spinal shock period.[6,7] Thereafter, transsynaptic degeneration due to a decrease in some form of neurotrophin became a popular theory.[8,9] Since then, there have been several studies related to this, but most studies have been conducted on patients with chronic SCI.[9,10] None have reported that peripheral nerve changes do not occur in all patients with SCI.

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The aim of this study was to assess the factors affecting electrophysiological changes of the peripheral nervous system in patients with subacute SCI. We compared the results of electrophysiological study of the lower extremities below the NLI and clinical information, including injury severity and functional scale, between groups with and without peripheral denervation potentials in the lower limb muscles below the NLI.

Methods

This study involved a retrospective review of medical records and was approved by the Institutional Review Board of Chungnam National University Hospital, Republic of Korea (IRB No. 2018-03-15). A total of 151 patients with SCI who underwent electrophysiological study to evaluate a myelopathy and peripheral nervous system in a rehabilitation center from January 2014 to June 2018 were initially enrolled in the study. Subjects were included if they (1) were older than 18 years; (2) were injured at a cervical or thoracic vertebrae level; and (3) had undergone electrophysiological study during the subacute phase (21-180d after the injury). Subjects were excluded if they had underlying conditions that had the potential to affect the results; these included diabetes mellitus, cancer, lower limb fracture, and radicular pain in the back and lower extremities previously. Patients taking medications known to cause peripheral neuropathy, those with severe lumbar foraminal stenosis or central canal stenosis, and those with insufficient clinical data were also strictly excluded from the study.[11,12]

For this research, subjects were divided into 2 groups, with or without the presence of denervation potentials on electromyography (EMG) below the NLI. Because direct injury of the peripheral nerve at or around the NLI is possible, we analyzed the EMG results of the lower limbs. The denervation potentials group had denervation potentials in 3 or more myotomes on bilateral lower limbs, and the no denervation potentials group had denervation potentials in 2 or less myotomes on uni- or bilateral limbs. Demographics and clinical characteristics such as sex, age, height, body mass index, etiology of injury, NLI, American Spinal Injury Association Impairment Scale (AIS), Korean version of Spinal Cord Independence Measure score, and Lower Extremity Motor Score were reviewed to identify factors influencing the presence or absence of denervation potentials. Other electrophysiological study results were compared between groups to identify the severity of peripheral nerve dysfunction.

Nerve conduction study

Dantec Keypoint G4[a] was used for the nerve conduction study. The nerve conduction study conducted in this laboratory was applied with the manual.[13] The peroneal nerve to the extensor digitorum brevis and the tibial nerve to the abductor hallucis were used to analyze motor nerve conduction, whereas the sural nerve was used for analysis of sensory nerve conduction. If necessary, to maintain skin temperature close to room temperature, the lower limb was warmed using a heat pack. The compound muscle action potential (CMAP) and sensory nerve action potential (SNAP) onset latencies were measured at the initial point of the negative phase of the potential. The CMAP amplitude was measured from baseline to the negative peak, and the SNAP amplitude was measured from the first negative peak to the following positive peak. The settings for the recordings of the motor studies were as follows: sensitivity, 3-5 mV/division; low frequency filter, 10 Hz; high frequency filter, 10 kHz; and sweep speed, 5 ms/division. The settings for the recordings of the sensory study were as follows: sensitivity of 20 μV/division; low frequency filter, 10 Hz; high frequency filter, 2 kHz; and sweep speed, 2 ms/division.

For the study of the peroneal nerves, subjects were in the supine position. The active electrode was placed on the extensor digitorum brevis, the reference electrode was placed on the fifth metatarsophalangeal joint, and the ground electrode was placed on the dorsal side of the foot. A distal nerve stimulation was applied 8 cm proximal to the active electrode, just lateral to the tibialis anterior tendon. A proximal nerve stimulation was applied just below the fibular head.

For the study of the tibial nerves, subjects were in the supine position. The active electrode was placed on the abductor hallucis, 1 cm behind and 1 cm below the navicular tubercle. The reference electrode was placed on the large toe. A distal nerve stimulation was applied with the cathode 8 cm proximal to the active electrode, posterior to the medial malleolus. A proximal nerve stimulation was applied to the crease of the popliteal fossa between the lateral third and the medial two-thirds.

For the study of the sural nerves, the active electrode was placed below and posterior to the lateral malleolus, and the reference electrode was placed 4 cm distally. Stimulation was applied approximately 14 cm proximal to the active electrode, slightly lateral to the midline, in the lower third of the calf.

Electromyography

Dantec Keypoint G4 was used for EMG. A monopolar Teflon-coated needle electrode was used together with a surface reference electrode and a ground electrode. The muscles evaluated in all subjects included at least 3 proximal muscles, namely, the tensor fasciae latae, iliopsoas, and vastus medialis, and 5 distal muscles, such as the peroneus longus, tibialis anterior, tibialis posterior, medial head of gastrocnemius, and abductor halluces, bilaterally. Precise needle placement was as described by Perotto et al.[14] The presence of abnormal spontaneous potentials such as fibrillations and positive sharp waves was noted.

Somatosensory-evoked potential

The Medelec Synergy 10 channel[b] was used for the somatosensory-evoked potential (SSEP) study with the guideline.[15] All participants were lying in a supine position with eyes closed and instructed to remain relaxed and quiet during testing. A bipolar bar electrode was used to stimulate the posterior tibial nerve between the posterior border of the medial malleolus and the medial border of the Achilles tendon. The cathode was placed close to the proximal end, and a monophasic square-wave electrical pulse of 0.1 ms was delivered by a constant current

List of abbreviations:

AIS	**American Spinal Injury Association Impairment Scale**
CMAP	**compound muscle action potential**
EMG	**electromyography**
MEP	**motor-evoked potential**
NLI	**neurologic level of injury**
SA	**spontaneous activity**
SCI	**spinal cord injury**
SNAP	**sensory nerve action potential**
SSEP	**somatosensory-evoked potential**

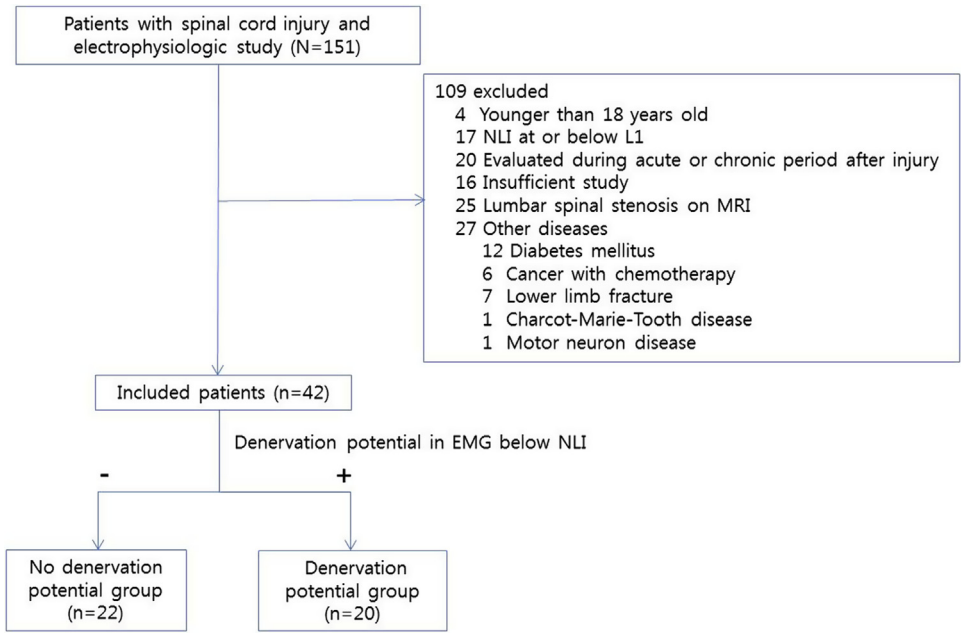

Fig 1 Flow diagram of the patient group classification.

stimulation, with a stimulation rate of 3 Hz. The intensity of stimulation was 2-3 times that of the sensory threshold. The active electrode was placed at Cz (P37), and FPz was used as a reference of the electroencephalogram International 10-20 system. The ground electrode was placed on the posterior neck around the C3 level. The electrode impedances between each electrode and the scalp were less than 5 KΩ, and the analysis time was 100 ms. We averaged at least 250 trials to ensure the reproducibility of cortical waves and measure their latency.

Motor-evoked potential

The MagPro stimulator with MagOption[c] was used for the motor-evoked potential (MEP) study with the guidline.[16] All participants

Table 1 Demographics and clinical characteristics of all patients with SCI/thoracic SCI

All SCI/Thoracic SCI	No Denervation Potential Group (n=22/4)	Denervation Potential Group (n=20/10)	*P* Value
Sex (n)			.432/.580
Male	14/3	15/5	
Female	8/1	5/5	
Age, mean ± SD (y)	59.59±12.21/68.75±3.40	56.50±12.34/56.50±12.04	.420/.013
Height, mean ± SD (m)	1.66±0.09/1.59±0.10	1.66±0.08/1.63±0.10	.933/.631
BMI, mean ± SD	22.59±2.72/23.36±1.87	22.08±3.92/23.91±3.88	.660/.736
NLI (n)			.031
Cervical myelopathy	18	10	
Thoracic myelopathy	4	10	
AIS (n)			<.001/.006
A	0/0	8/4	
B	0/0	2/2	
C	4/0	10/4	
D	18/4	0/0	
Korean version of spinal cord independence measure, mean ± SD	50.81±31.75/76.25±17.65	21.75±16.64/33.80±13.84	.001/.009
lower extremity motor score, mean ± SD	33.64±13.60/32.80±5.89	4.35±7.74/4.50±7.83	<.001/<.001
Average time from injury to electrophysiological study, mean ± SD (d)	58.64±34.09/63.25±23.31	66.45±42.16/75.60±46.74	.243/.526
Etiology (n)			.832/.580
Traumatic injury	5/2	4/7	
Nontraumatic injury	17/2	16/3	

Abbreviation: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

Table 2 Results of electrophysiological study of all patients with SCI

Variables	No Denervation Potential Group (n=44)	Denervation Potential Group (n=40)	*P* Value
NCS			
Peroneal nerve			
Absent (n)	5	24	<.001
Induced (n)	39	16	
Latency, mean ± SD (ms)	3.90±0.86	4.11±0.84	.409
Amplitude, mean ± SD (mV)	2.78±1.82	1.43±1.32	.007
Tibial nerve			
Absent (n)	1	14	<.001
Induced (n)	43	26	
Latency, mean ± SD (ms)	3.40±0.84	4.24±1.07	.001
Amplitude, mean ± SD (mV)	9.50±4.02	4.02±2.67	<.001
Sural nerve			
Absent (n)	2	6	.105
Induced (n)	42	34	
Latency, mean ± SD (ms)	2.57±0.46	2.63±0.54	.594
Amplitude, mean ± SD (μV)	9.74±4.60	6.20±3.01	<.001
SSEP			
Absent (n)	8	32	<.001
Evoked (n)	36	8	
Latency, mean ± SD (ms)	44.667.84	45.48±4.82	.645
MEP			
Absent (n)	10	35	<.001
Evoked (n)	34	5	

Abbreviation: NCS, nerve conduction study.

were instructed to relax their limbs in a seated position. Surface electrodes were placed bilaterally on the skin, over the muscle belly of the tibialis anterior. Magnetic stimulation was applied tangentially over the area corresponding to Cz, through a circular coil, with 65% of the stimulation output. The evoked potential was recorded with a Dantec Keypoint EMG machine. A positive response was defined as the presence of a signal >50 μV. The MEP of the "absent" group showed no positive response from 65% stimulation, with stimuli delivered at least 5 times to confirm this.

Statistical analysis

The collected data were statistically analyzed using SPSS version 23.0.[d] Depending on data number, chi-square test or Fisher exact test was used to analyze categorical data. For continuous data, depending on the normality, either a *t* test or Mann-Whitney *U* test was used for analysis. Statistical significance was defined as $P<.05$.

Results

Of the 151 subjects, 109 were excluded for the following reasons: 4 were younger than 18 years; 17 were injured at or below L1; 20 had not undergone an EPS between 21 and 180 days after the injury; 25 had severe spinal stenosis; and 27 had diseases that caused peripheral neuropathy. A total of 42 patients were finally selected and separated into the no denervation potential group (n=22) and the denervation potentials group (n=20) (fig 1).

There were no significant differences in sex, age, height, and body mass index between the no denervation potentials group and the denervation potentials group. The denervation potentials group showed a higher proportion of thoracic myelopathy ($P=.031$) and more severe SCI than the no denervation potentials group ($P<.001$). The functional indicators, Korean version of spinal cord independence measure and lower extremity motor score, were significantly lower in the denervation potentials group ($P<.001$). The average time from injury to EPS was longer in the denervation potentials group, although this was not statistically significant. The etiology of injury was also not statically different between the groups (table 1).

For analysis of the electrophysiological study, we used the results of lower extremities from both sides. All 44 results of the electrophysiological study in the no denervation potentials group and 40 results in the denervation potentials group were used for analysis. In the peroneal nerve study, there was a significantly high prevalence of no response and a lower amplitude in the denervation potentials group than in the no denervation potentials group ($P<.001$). In the tibial nerve study, there was a significantly high prevalence of no response with a more delayed latency and lower amplitude in the denervation potentials group than in the no denervation potentials group ($P<.001$). The sural nerve study showed a significantly lower amplitude in the denervation potentials group ($P<.001$) compared than the no denervation potentials group. In both the SSEP and MEP studies, there was a significantly higher prevalence of no response in the denervation potentials group ($P<.001$) (table 2).

Because the proportion of thoracic myelopathy was significantly higher in the denervation potentials group than in the no denervation potentials group, subgroup analysis was conducted. In the subgroup analysis of thoracic SCI, there was no significant

Table 3 Results of electrophysiological study of patients with thoracic SCI

Variables	No Denervation Potential Group (n=8)	Denervation Potential Group (n=20)	*P* Value
NCS			
Peroneal nerve			
Absent (n)	1	14	.011
Evoked (n)	7	6	
Latency, mean ± SD (ms)	3.45±0.67	3.64±0.72	.628
Amplitude, mean ± SD (mV)	3.56±1.72	2.71±1.09	.305
Tibial nerve			
Absent (n)	0	8	.038
Evoked (n)	8	12	
Latency, mean ± SD (ms)	3.18±0.72	4.02±0.93	.034
Amplitude, mean ± SD (mV)	9.87±3.08	4.27±2.61	.001
Sural nerve			
Absent (n)	0	0	
Evoked (n)	8	20	
Latency, mean ± SD (ms)	2.71±0.29	2.51±0.41	.167
Amplitude, mean ± SD (μV)	6.11±3.45	6.45±3.46	.817
SSEP			
Absent (n)	1	14	.007
Evoked (n)	7	6	
Latency, mean ± SD (ms)	45.06±4.89	43.78±3.63	.945
MEP			
Absent (n)	0	16	<.001
Evoked (n)	8	4	

Abbreviation: NCS, nerve conduction study.

difference in sex, age, height, body mass index, or NLI between the no denervation potentials group and the denervation potentials group. However, the denervation potentials group was significantly younger ($P=.013$) with more severe SCI ($P=.006$). The functional indicators were also significantly lower in the denervation potentials group ($P=.009$). The average time from injury to EPS and the etiology of injury was not significantly different between each group (see table 1).

In the peroneal nerve study of thoracic SCI, there was a significantly high prevalence of no response in the denervation potentials group ($P=.011$). However, the latency and amplitude of CMAP were not significantly different. In the tibial nerve study, there was a significantly higher prevalence of no response and a more delayed latency in the denervation potentials group than in the no denervation potentials group ($P=.038$). The denervation potentials group also showed lower amplitude than the no denervation potentials group ($P=.001$). The sural nerve study showed no significant difference between the 2 groups. In the SSEP and MEP studies, there was a significantly high prevalence of no response in the denervation potentials group ($P<.001$) (table 3).

Discussion

This study applied strict criteria for the selection of subjects to exclude any factors that may have affected the peripheral nervous system. In the analysis of demographics and clinical characteristics, the denervation potentials group showed more severe SCI based on the AIS, lower extremity motor score, and Korean version of spinal cord independence measure than did patients without denervation potentials. There were no significant differences in sex, age, height, and body mass index between the 2 groups. In the analysis of electrophysiological study, the denervation potentials group showed a higher proportion of no response than no denervation potentials group. Furthermore, there were significantly lower amplitudes of motor and sensory nerve conduction in the denervation potentials group than in the no denervation potentials group.

The thoracic SCI showed a high incidence of peripheral nerve abnormalities; however, subgroup analysis showed the proportion of severe SCI patients, especially AIS A and B, was high in the thoracic SCI group. In the AIS C group, there were no significant differences in demographics and clinical characteristics or in the results of electrophysiological study between the denervation potentials group and the no denervation potentials group (data not shown). We consider that the degree of injury corresponding to grade C has a broader spectrum than other grades according to the International Standards for Neurological Classification of SCI.

In previous studies, the result of EPS above NLI has been normative.[17] Thus, there is a possibility that disused muscle fibers may have an effect on the motor conduction study. However, the influence of muscle atrophy on CMAP amplitude is likely minor. Indeed, in a previous study, after 2 weeks of immobilization of the healthy participants' limb without muscle atrophy in an experimental condition, there was significant muscle atrophy with a 22% reduction in maximum muscle power, but the electrophysiological results remained unchanged.[18]

In this study, the motor nerve conduction latency was not significantly different between the denervation potentials group and the no denervation potentials group. Despite this, the amplitude was largely decreased, and there were many cases that were not induced in the denervation potentials group. This

suggests that there was increased axonal loss, which is caused by death or dysfunction of cell body, in the denervation potentials group compared with the no denervation potentials group. The most popular theory for the cause of motor neuron dysfunction is the transsynaptic degeneration of motor neurons whereby the activity of the second-order motor neuron is depressed after injury of the central nervous system.[8-10,19,20] In a previous study, dysfunction of anterior horn cells resulted in an barrier at the neuromuscular junction as a result of failure of axonal flow and degeneration of the axons; this in turn leads to a decreased CMAP amplitude.[21] In addition, previous studies have reported histologic changes in motor neurons, including a reduction in the number of dendrites per motor neuron, a decrease in dendritic size, and attenuation of dendritic branching.[22] Such changes are thought to be due to a decrease in the levels of a survival-promoting neurotrophic molecule, neurotrophin, exchanged via synapses. Previous studies have also shown that SCI diminishes the transcription of neurotrophins, such as brain-derived neurotrophic factor, neurotrphin-3, and neurotrophin-4. Neurotrophin promotes neuron survival, modulates neuroplasticity, and is essential in maintaining and enhancing the function of neuromuscular junction.[23-25] In a previous study using an animal SCI model, it was SAs were considerably improved in muscles below the NLI when fibroblasts, secreting brain-derived neurotrophic factor and neurotrphin-3, were grafted to the lesion of the SCI.[26] We consider that the severity of SCI accelerates the transsynaptic degeneration and at least partially explains the difference between the denervation potentials group and the no denervation potentials group.

In the current study, the sensory nerve conduction latency was not significantly different between the denervation potentials group and the no denervation potentials group. However, the amplitude was significantly decreased, and there were more cases that were not induced in the denervation potentials group. This is similar to the result of a previous study that demonstrated that 75% of patients with chronic complete SCI lacked an SNAP in the sural nerve.[3] Transsynaptic degeneration seems to affect not only the motor neurons but also the dorsal root ganglion. It can be inferred from a previous study that a change in gene expression of the dorsal root ganglion is induced after cord stimulation using an animal SCI model, suggesting that the descending signal through the spinal cord also affects the sensory nerve.[27]

Furthermore, a previous study demonstrated that the abnormal results of electrophysiological study can occur as a result of changes in the tissues of the limbs below the lesion after SCI, for example, impaired blood circulation in paralyzed extremities. This functions to lower the temperature of the muscles and the peripheral nerve, delaying the electrophysiological process.[28] In this study, although the temperature of the skin was not controlled exactly in all cases, we made every effort to maintain the skin temperature close to room temperature. Thus, it can be considered that the effect of temperature was minimal.

Study limitations

Given the retrospective nature of this study, it was limited to the observation of changes in the results of the EPS while the patients with SCI progressed from the subacute phase to the chronic phase. In addition, to identify how the dysfunction in the peripheral nervous system affects the functional prognosis of patients with SCI, it is necessary to follow functional recovery of each grade of the AIS. However, this was not possible because of the small number of patients in the subgroup and the low follow-up rate. Therefore, we consider that further study is necessary to validate these findings.

Conclusion

Through this study, we were able to confirm that severe SCI, based on AIS, can induce electrophysiological changes of the peripheral nervous system as well as the central nervous system in subacute SCI. In severe subacute SCI, axonal loss was detected in peripheral nerves, including both motor and sensory nerves below NLI, and this is explained by transsynaptic degeneration.

Suppliers

a. Dantec Keypoint G4; Neurolite.
b. Medelec Synergy 10 channel; VIASYS Healthcare.
c. MagPro stimulator with MagOption; MagVenture.
d. SPSS version 23.0; IBM.

Keyword

Rehabilitation

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